

Summary for Jul 18

Risk-Free Price:

Short-Rate Process: Cox–Ingersoll–Ross (CIR)

$$dr_t = \kappa_r(\theta_r - r_t) dt + \sigma_r \sqrt{r_t} dZ_t^r.$$

- **Motivation:**

- Motivated by James’s proposal
- Ensures nonnegative nominal rates, matching real bond markets.
- Mean-reversion toward a central-bank target θ_r .
- State-dependent volatility: calm when rates are low, turbulent when high.

- **Term/Parameter Definitions:**

- r_t : instantaneous short-term interest rate at time t
- dr_t : change in the short-term interest rate.
- $\theta_r > 0$: long-run average level of the short rate.
- Z_t^r : standard Brownian motion driving for rate shocks.
- $\kappa_r(\theta_r - r_t) dt$: deterministic drift pulling r_t toward θ_r .
 - * $\kappa_r > 0$: speed of mean reversion to θ_r (how quickly r_t is pulled back toward θ_r).
 - * $(\theta_r - r_t)$: distance from long-run mean.
 - * dt : infinitesimal time increment.
- $\sigma_r \sqrt{r_t} dZ_t^r$: stochastic shock to r_t .
 - * $\sigma_r > 0$: volatility coefficient—scales the size of random shocks.
 - * σ_r : scales volatility of shocks.
 - * $\sqrt{r_t}$: state-dependent volatility—larger when r_t is high.
 - * dZ_t^r : increment of standard Brownian motion.

- **Key Features:**

- **Nonnegativity:** If

$$2\kappa_r \theta_r \geq \sigma_r^2 \quad (\text{the Feller condition}),$$

then r_t stays strictly positive.

- **State-Dependent Volatility:** The diffusion term $\sigma_r \sqrt{r_t}$ shrinks when rates are low, matching empirical calm in low-rate regimes.
- **Analytic Bond Pricing:** Under CIR, zero-coupon bond prices have known closed-form expressions involving confluent hypergeometric functions.

- **Advantage:** By ensuring $r_t \geq 0$ and giving volatility proportional to $\sqrt{r_t}$, CIR captures two realistic phenomena—no negative nominal rates and lower turbulence when rates are near zero—while still admitting semi-analytic solutions.

Risky Asset Price

Notation: we combine the formulation of Heston-Style with the Stochastic Drift and Volatility Model by combining the SDE of drift term with the Heston-Style SDE, as you can have a reference for in the below.

Risky Asset Price: Heston-Style

$$\frac{dS_t}{S_t} = \mu_S dt + \sqrt{v_t} dW_t^S.$$

$$d\mu_t = \kappa_\mu(\theta_\mu - \mu_t) dt + \xi_\mu dW_t^{(1)}$$

$$dv_t = \kappa_v(\theta_v - v_t) dt + \sigma_v \sqrt{v_t} dZ_t^v.$$

– **Motivation:**

- * To reflect that returns' volatility changes over time (volatility-of-volatility), not just the price level.

– **Term/Parameter Definitions:**

- * S_t : risky asset price.
- * dS_t/S_t : instantaneous return on the risky asset.
- * μ_S : expected return/ constant or slowly-varying risk premium.
- * $\mu_S dt$: expected drift component.
- * v_t : instantaneous variance.
- * W_t^S : Brownian motion for returns. Correlated with Z_t^v .
- * $\sqrt{v_t} dW_t^S$: random return fluctuation.
 - $\sqrt{v_t}$: time-varying volatility level.
 - dW_t^S : Brownian motion increment for returns.

– **Key Features:**

- * Volatility clustering via v_t ./ Instantaneous volatility $\sqrt{v_t}$ captures regime shifts in market risk.
- * Flexibility to model skew and kurtosis./ Correlation between Z_t^v and W_t^S can generate leverage effects.

- **Advantage:** Integrates stochastic variance into portfolio choice, enriching risk dynamics. Reproduce empirically observed return skewness and kurtosis, improving calibration to real markets and better capturing tail-risk.

• Below Are For References

1. Variance Process: Square-Root (Heston-Style)

$$dv_t = \kappa_v(\theta_v - v_t) dt + \sigma_v \sqrt{v_t} dZ_t^v.$$

– **Motivation:**

- * To model volatility clustering—periods of high market turbulence followed by calm—and maintain nonnegative instantaneous variance.

– **Term/Parameter Definitions:**

- * v_t : instantaneous variance of the risky asset return.
- * dv_t : change in instantaneous variance of the risky asset.
- * $\kappa_v > 0$: speed of mean reversion for variance.
- * $\theta_v > 0$: long-run average variance.
- * $\sigma_v > 0$: volatility of volatility, scaling random shocks' amplitude.
- * Z_t^v : Brownian motion (possibly correlated with W_t^S).
- * $\kappa_v(\theta_v - v_t) dt$: mean-reverting drift of variance.
- * $\sigma_v \sqrt{v_t} dZ_t^v$: stochastic volatility shocks.
- * $\sqrt{v_t}$: ensures variance stays nonnegative.

* dZ_t^v : Brownian increment.

- **Key Features:**

- **Nonnegativity:** Same Feller-type condition

$$2\kappa_v\theta_v \geq \sigma_v^2$$

keeps $v_t \geq 0$.

- **Volatility Clustering:** High v_t begets more randomness in returns, which then mean-reverts, capturing clustering.
- **Flexibility:** Allows skewed and heavy-tailed return distributions when plugged into a Heston-style price process.
- **Advantage:** Provides realistic stochastic volatility with well-known calibration procedures. This process gives a far more realistic model of equity returns—volatility is itself random and mean-reverting—while retaining tractability for simulation and, in some cases, semi-analytic option-pricing formulas.

2. Risky Asset

2.1 Stochastic Drift and Volatility Model

$$dP_1(t) = P_1(t) [\mu_t dt + \sigma_t dZ_t]$$

where both the **drift** μ_t and the **volatility** σ_t evolve according to their own stochastic differential equations:

$$\begin{aligned} d\mu_t &= \kappa_\mu(\theta_\mu - \mu_t) dt + \xi_\mu dW_t^{(1)} \\ d\sigma_t &= \kappa_\sigma(\theta_\sigma - \sigma_t) dt + \xi_\sigma dW_t^{(2)} \end{aligned}$$

- $P_1(t)$: Price of the risky asset at time t .
- μ_t : Instantaneous expected return (drift), governed by a mean-reverting process.
- σ_t : Instantaneous volatility, also governed by a mean-reverting process.
- dZ_t : Standard Brownian motion driving asset price shocks.
- $dW_t^{(1)}, dW_t^{(2)}$: Independent Brownian motions driving shocks to μ_t and σ_t , respectively.
- $\kappa_\mu, \kappa_\sigma$: Speeds of mean reversion for μ_t and σ_t .
- $\theta_\mu, \theta_\sigma$: Long-run mean levels of drift and volatility.
- ξ_μ, ξ_σ : Volatilities of the drift and volatility processes.

2.2 Key Features

Feature	Description
Stochastic Drift	The drift μ_t fluctuates around a long-term mean, capturing cyclical excess returns or macroeconomic shocks.
Stochastic Volatility	The volatility σ_t evolves dynamically, modeling realistic volatility clustering.
Mean-Reversion	Both μ_t and σ_t revert to their respective long-run means.
Multiple Shocks	Separate Brownian motions for drift and volatility allow for rich, multidimensional randomness.
Financial Realism	Suitable for capturing momentum, regime-switching, or liquidity-driven effects.

TBD: Contribution Benchmark: Feedback Rule

$$dM_t = [c_0 + \eta(W_t - M_t)] dt.$$

- **Motivation:**
 - * To endogenize savings behavior: investors contribute more when they're ahead and less when they're behind.
- **Term/Parameter Definitions:**
 - * M_t : cumulative net contributions.
 - * $c_0 \geq 0$: base contribution rate.
 - * η : sensitivity of contributions to wealth–benchmark gap.
 - * W_t : current Wealth
 - * dM_t : change in cumulative contributions.
 - * $c_0 dt$: baseline contribution flow.
 - * $\eta(W_t - M_t) dt$: performance-driven adjustment.
 - $(W_t - M_t)$: current surplus (if positive) or shortfall (if negative).
- **Key Features:**
 - * When $M_t > W_t$, contributions rise; if behind, contributions fall.
 - * Captures effects in saving, linking regret to saving rates.
- **Advantage:** Captures behavioral saving patterns and enriches regret dynamics.

TBD: Wealth Dynamics

$$dW_t = \left[r_t(1 - \alpha_t)W_t + \mu_S \alpha_t W_t - (c_0 + \eta(W_t - M_t)) \right] dt + \sqrt{v_t} \alpha_t W_t dW_t^S.$$

- **Motivation:** Combine all sources of return and saving into a self-financing framework.
- **Term/Parameter Definitions:**
 - * $\alpha_t \in [0, 1]$: fraction of wealth invested in S_t .
 - * $r_t(1 - \alpha_t)W_t$: risk-free drift.
 - * $r_t(1 - \alpha_t)W_t dt$: return from risk-free portion.
 - * $\mu_S \alpha_t W_t$: equity expected drift.
 - * $\mu_S \alpha_t W_t dt$: expected return from risky portion.
 - * $-(c_0 + \eta(W_t - M_t)) dt$: net contributions flow out of wealth.
 - * $\sqrt{v_t} \alpha_t W_t dW_t^S$: stochastic equity return.
- **Key Features:**
 - * **Multi-Source Risk:** Wealth is influenced by rate risk, equity risk, and volatility-of-volatility.
 - * **Endogenous Saving:** Contributions shift with performance.
 - * **Behavioral Feedbacks:** Falling behind the benchmark reduces contributions, potentially exacerbating regret.
- **Advantage:** Although high-dimensional, this SDE captures rich interactions between interest-rate risk, equity risk, saving behavior, and regret. I believe It could provide a solid foundation work and formulation for how these multiple frictions shape optimal dynamic portfolio and saving decisions. However, we might need requiring more advanced numerical methods(e.g. high-dimensional PDEs or Monte Carlo).